

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 23

Accessibility

Def: State j is said to be accessible from state i if there exists $n \geq 0$ such that $p_{ij}^{(n)} > 0$, where $p_{ij}^{(0)} = \delta_{ij}$.

Notation: $P_i(\cdot) = P(\cdot | X_0 = i)$

Equivalent Conditions

Theorem: For distinct states i and j , the following are equivalent.

- ① j is accessible from i .
- ② $P(\text{Ever be in } j | \text{starting from } i) > 0$.
- ③ $p_{i_0, i_1} p_{i_1, i_2} \cdots p_{i_{n-1}, i_n} > 0$ for some states $i_0 = i, i_1, i_2, \dots, i_n = j$ and for some $n \geq 0$.

Proof: (Equivalence of 1 and 2)

$$\begin{aligned} p_{ij}^{(n)} &\leq P_i(\cup_{n=0}^{\infty} \{X_n = j\}) \\ &= P(\text{Ever be in } j | \text{starting from } i) \\ &= P_i(X_n = j \text{ for some } n \geq 0) \\ &\leq \sum_{n=0}^{\infty} p_{ij}^{(n)}. \end{aligned}$$

This proves the equivalence of 1 and 2.

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Proof: (Equivalence of 1 and 3)

From the definition of matrix multiplication,

$$p_{ij}^{(n)} = \sum_{i_1, i_2, \dots, i_{n-1} \in S} p_{i, i_1} p_{i_1, i_2} \cdots p_{i_{n-1}, j}.$$

This proves the equivalence of 1 and 3.

Communication

Def: Two states i and j are said to communicate if i and j are accessible from each other, i.e., there exist $m \geq 0$ and $n \geq 0$ such that $p_{ij}^{(n)} > 0$ and $p_{ji}^{(m)} > 0$.

Notation: $i \rightarrow j$: j is accessible from i .
 $i \leftrightarrow j$: i and j communicate.

- Property of communication:

① (Reflexivity) $i \leftrightarrow i$.

② (Symmetry) $i \leftrightarrow j \iff j \leftrightarrow i$.

③ (Transitivity) $i \leftrightarrow k$ and $k \leftrightarrow j \implies i \leftrightarrow j$.

- Thus, communication is an equivalence relation. Hence, it partitions the state space into communicating classes.

Closed Class and Irreducibility

Def: A communicating class C is said to be closed if

$$i \in C, i \rightarrow j \text{ imply } j \in C.$$

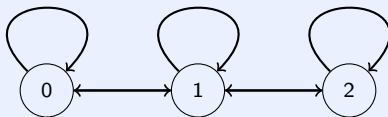
Thus, a closed class is one from which you cannot go out.

Def: A state i is called absorbing if $\{i\}$ is a closed class.

Def: A MC is said to be irreducible if all states communicate with each other, *i.e.*, there is a single communicating class.

Example

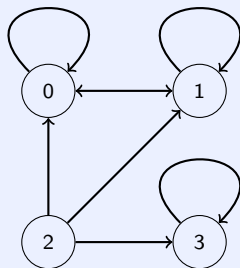
Example 1: $P_1 = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/4 & 1/4 \\ 0 & 1/3 & 2/3 \end{pmatrix}$



- $\{0, 1, 2\}$ is a closed communicating class.
- It is an irreducible MC.

Example

Example 2: $P_2 = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

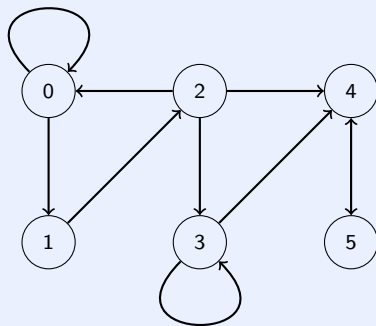


- $\{0, 1\}$, $\{2\}$ and $\{3\}$ are communicating classes.
- $\{0, 1\}$ and $\{3\}$ are closed, $\{2\}$ is not closed.
- It is not an irreducible MC.

Example

Example 3: $P_3 =$

$$\begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1/6 & 0 & 0 & 1/3 & 1/2 & 0 \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$



- $\{0, 1, 2\}$, $\{3\}$ and $\{4, 5\}$ are communicating classes.
- $\{4, 5, \}$ is closed.
- It is not an irreducible MC.